

40-Bragg: Measuring Work Function of Non-Metallic Materials Using KPFM

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Abstract:

The work functions of twelve pre-prepared polymer films were measured using Kelvin Probe Force Microscopy with a Bruker Dimension 3100 using SCM-PIT-V2 tips the values of the work functions ranged from $3.72 \pm 0.08 \text{ eV}$ to $6.1 \pm 0.13 \text{ eV}$. The largest source of uncertainty in these measurements was from the tip calibration

An attempt was made to study the effect of polymer film thickness on the work function by fabricating new PVC films by spin coating multiple layers onto the same sample pucks. Measurements of film thickness on these samples were unsuccessful, preventing further study at this time. Heat pressing layers of nylon film onto substrates was also attempted but similarly, yielded unsuccessful results.

PVC was spin coated onto steel and gold-plated pucks to observe any effects of substrate material on work function measurements. A difference of $0.2 \pm 0.1 \text{ eV}$ was measured, matching the difference in work function of the two substrates, showing that the substrate material was affecting the measurements. Further measurements and samples will be needed to confirm this.

The design of an existing Kelvin Probe was improved, adding a better sample stage and a larger, gold-plated tip as well as some ergonomic and functional improvements for ease of use.

Sample Preparation:

Spin Coating:

Methods:

15mm diameter steel pucks polished with $1 \mu\text{m}$, $0.3 \mu\text{m}$ and $0.05 \mu\text{m}$ grit for a smooth finish.

0.2g high molecular weight Poly(vinyl chloride) from Sigma-Aldrich added to 5ml acetone and sonicated. After two days the undissolved PVC was removed from the solution.

Steel pucks spun at 2800rpm, $20 \mu\text{l}$ of solution deposited onto already spinning puck using a Hamilton syringe. Puck spun for 60 seconds, left to dry for 5 minutes before having another layer added. 1-5 layers of solution were added to the pucks.

The thickness of the films was measured using a Bruker DekTakXT by cutting a line into the film using a scalpel.

Results:

Thickness measurements of the films using the DekTak were unsuccessful, yielding surface profiles that the film thickness could not be determined from.

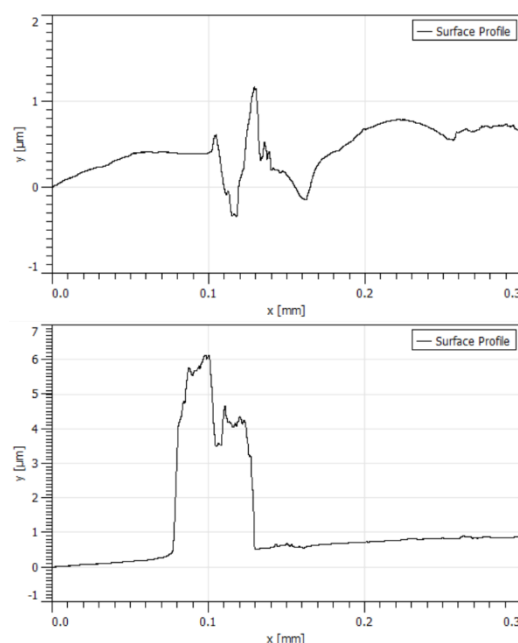


Figure 1:

Surface profile measurements of 1 layer (top) and 5 layers (bottom). Data is very poor and thicknesses cannot be determined

Discussion:

Initially, the solution was too viscous, leading to poor coverage of the puck. 1ml of acetone was added to the solution to decrease the concentration and viscosity, enabling a more even coating.

Environmental factors such as temperature, humidity and airflow strongly influence the thickness of the film.

Manually using a Hamilton syringe to deposit the solution led to the solution not being deposited centrally on the pucks, not having an even flow speed and some splashing, causing the films to be a lower quality than desired.

An automatic dispenser could be used to deposit the solution to improve the film quality.

Visually, under a microscope the addition of more layers caused the surface to become rougher. Images of this can be found in the appendix on page 9 (figure 8).

Issues with thickness measurements could be attributed to errors when cutting the film. The substrate may have been cut as well as the film, causing poor measurements or the film may not have been cut, although a cut was clearly visible on the sample.

Placing something on the substrate to mask part of the sample during spin coating may be a better way to be able to measure the thickness.

Heat Pressing:

Methods:

17 μ m films of nylon were heat pressed onto steel, aluminium and copper substrates, g-clamped to a hotplate at 200°C for 10 minutes.

Results:

No surface adhesion was achieved with any of the samples apart from potentially one aluminium sample, however this needs verifying.

Discussion:

This poor result may be attributed to incorrect methods. More attempts using different methods and materials are required to determine whether this method of sample preparation would work and be useful. The one successful heat press was done using two separate heats, one at a lower temperature and lower pressure and one at a higher temperature

and higher pressure. After this a film was removed from the Aluminium surface leaving behind a residual film. After scanning the film, a potential measurement of 4.3eV was recorded, which differs from the aluminium work function of 4.17eV (1) used for calibration. This implies the nylon is exerting some effect on the reading, however it is not clear if it is behaving as a dielectric or as a sample surface. It is more likely to be behaving as a dielectric as nylon is higher than aluminium on the triboelectric series so a work function lower than 4.3eV would be expected for a surface sample.

It is important not to use a g-clamp on a ceramic hot plate without pre-heating as the plate may break. This can be expensive to fix and be very irritating.

It is important to note that the nylon sheet in the project student box has two layers, one nylon and one not. We do not know which is which.

Kelvin Probe Force Microscopy:

Pre-Prepared Polymer films:

Methods:

A range of 12 pre-prepared polymer films were provided. The work functions of these films were measured using a Bruker D3100, with an SCM-PIT-V2 tip, calibrated with an Au/Al calibration grid from Budget Sensors. Instructions on setting up the D3100 for KPFM can be found in the appendix on page 8.

Measurements were repeated over multiple days and areas of the films and averaged for the results.

Results:

A full table of results can be found in the appendix on page 8 (table 2)

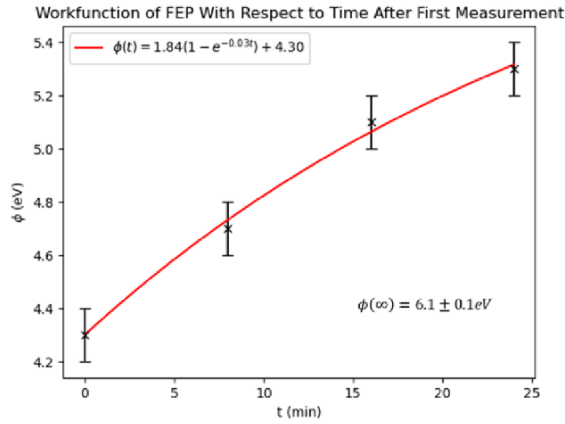


Figure 2:
Work function of FEP vs time with exponential saturation curve fitted. The curve plateaus at $6.1 \pm 0.1 \text{ eV}$, matching the expected value of $6.03 \pm 0.05 \text{ eV}$ (2)

Discussion:

Some polymers provided could not be measured due to separation of the polymer layer from the substrate, causing bouncing of the AFM tip.

During calibration it was noticed that on the strips of gold/aluminium, the CPD did not plateau, giving incorrect tip calibrations. To avoid this, calibration was done on the interface between the gold and aluminium on the calibration sample shown below.

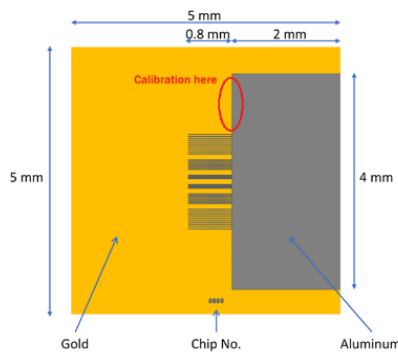


Figure 3:
Schematic diagram of calibration grid from Budget Sensors (3)

Values of 4.83 eV (4) and 4.17 eV were used as the work functions for gold and aluminium respectively during calibration.

Figure 2 shows electric discharge of the FEP sample. It is important to check for charging in the samples to ensure reliable measurements. In the future, the charging properties of the different polymers can be studied.

It is also important to repeat measurements over different days and using different KPFM tips to ensure that results are reproducible.

Highly Ordered Pyrolytic Graphite:

Methods:

Highly ordered pyrolytic graphite (HOPG) can be used for calibration.

A sample of HOPG was provided and subsequently measured using KPFM on the Bruker D3100.

Results:

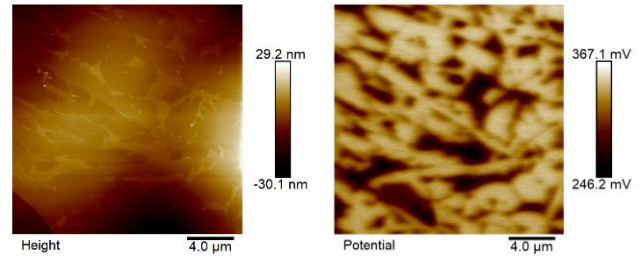


Figure 4:
Topography (right) and Surface potential (left) of HOPG

Between layers:

$$\Delta_{CPD} = 80 \text{ mV} \quad \Delta_h = 2.6 \text{ nm}$$

Discussion

This can be used for calibration. We did not use it

Effects of Substrate Material:

The variations in substrate work function caused a corresponding change in film work function. This can be explained by these two spins coated samples had poor entrainment to the surface. This is supported by the extreme variation in topography.

Methods:

150nm of Au evaporated onto smooth steel puck.

Work function of gold coated puck and steel puck measured.

Steel and gold coated puck spin coated with one layer of acetone/PVC solution using the procedure described in the Spin Coating section within Sample Preparation on page 2. The thickness of the films and their work functions were then measured

Results:

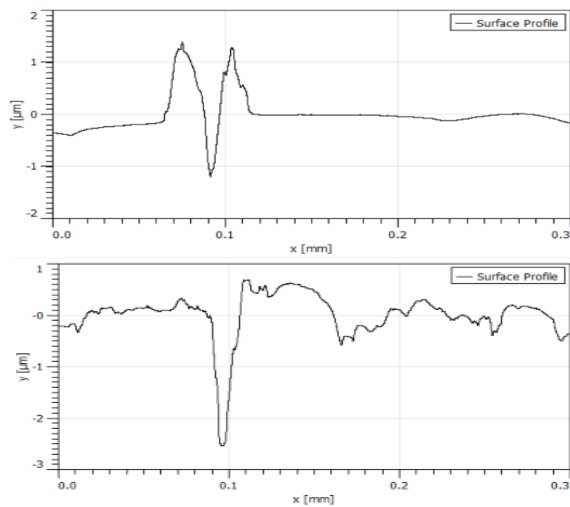


Figure 5:

Surface profile of PVC on Steel (left) and on Gold (right)

Sample	Thickness (μm)	Work Function (eV)	Uncertainty (eV)
Steel	-	4.8	0.1
Gold	-	5.0	0.1
Difference	-	0.2	0.1
PVC on steel	1.1 ± 0.3	5.1	0.1
PVC on gold	2.8 ± 0.2	5.3	0.1
Difference	1.7 ± 0.5	0.2	0.1

Table 1:

Thickness and work functions of samples

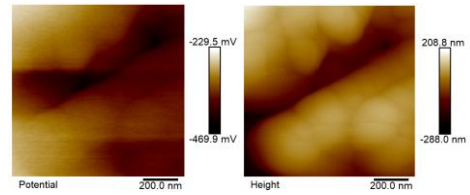


Figure 6:

Topography (right) and Surface potential (left)
PVC on gold

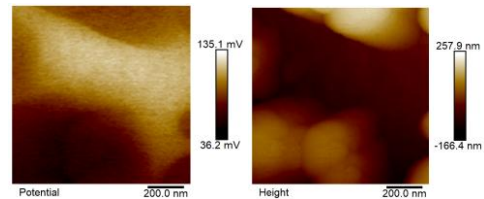


Figure 7:

Topography (right) and Surface potential (left)
PVC on steel

Discussion:

This data demonstrates a correlation between film substrate and surface work function. However, the films thickness varies in height by 400nm-500nm for both gold and steel, which is significantly higher than their bare surfaces (figure 7). There is also significantly higher variation in surface potential, when compared to other PVC samples. This would suggest a poor coating adhesion, rather than a clear relationship between sample substrate and surface potential. Further tests using the remaining gold pucks will be necessary.

It is interesting to note that the gold-plated substrate, which is rougher than the steel substrate gave a thicker film by almost $2\mu\text{m}$.

The difference in work function on the polymer samples of $0.2 \pm 0.1\text{eV}$ is the same as the difference of the substrates, indicating that, at this film thickness, the substrate does affect the samples work function.

Effect of film thickness:

During this project we have been unable to determine the effects of film thickness due to the lack of time and hinderances with sample preparation.

Methods:

Polished steel pucks were spin coated with layers of the acetone/PVC solution using the procedure described in the Spin Coating section within Sample Preparation on page 2. Once coated, the pucks were left to air dry in a fume hood for five minutes before having another layer added.

This was done until we had samples ranging from 1-5 layers.

If this were to be done in the future, rather than spin coating multiple layers, the parameters for spin coating could be changed following the equation

$$h = k_4 C_0 (\eta_0 \Delta_0)^{\beta'} \omega^\alpha \quad (\text{Equation 1})$$

Where h is the film thickness, k_4 is a constant, C_0 is the initial polymer concentration, η_0 is the initial solution viscosity, Δ_0 is the solute diffusivity, β' and α are independent exponents usually equal to 0.25 and -0.5 respectively and ω is the angular velocity of the substrate during coating. (5)

This equation can be simplified to

$$h(\omega) = k_1 \omega^\alpha \quad (\text{Equation 2})$$

to only show the effect of spin speed.

Results:

No useable results were acquired.

Discussion:

The thickness of these samples was unable to be determined using the Bruker DekTakXT.

Further discussion can be found in the discussion section for spin coating within the Sample Preparation section on page 2.

In the future, different methods of manufacturing films of different thickness should be used such as changing the spin speed or concentration in order to observe the effects of thickness and fermi level alignment. An AFM may be able to be used to measure the thicknesses of the films, as it has a smaller tip diameter, giving more accurate results, but may struggle with thicker films as it may not be able to travel deep enough to correctly measure the film thickness.

Kelvin Probe:

Standard operation:

The Kelvin probe's purpose is to measure the CPD of a sample in much the same way as KPFM. An Oscillating probe creates a capacitor between itself and the sample, which is grounded. A DC bias can be applied to the tip, using a pre-amp, to demonstrate a linear relationship between the measured CPD and bias. The point at which bias is zero on this line is the non-calibrated work function. To calibrate the kelvin probe, the previous steps can be repeated using a clean gold puck, with a known work function. The mechanical oscillation of the probe is done by a piezo, and this generates a measurable AC signal response, which is measured using a lock in amp. A circuit diagram of this can be found on page 11 of the appendix (figure 11). To achieve a strong signal, the sample and probe should be close together, which is done by using a 3-axis stage. The readings are taken within a faraday box to limit the interference from background sources.

Design problems and upgrades:

The central problem with this Kelvin probe design is a lack of usability, which manifests in four keyways; loose pre-amp, fragile tip, difficult manual stage.

The loose pre-amp negatively affects the design, as it attaches to the probe via a coaxial cable. This means that if a user knocks the pre-amp during set up, the fragile probe will become misaligned with the sample, which will invalidate any previous calibration, whilst also reducing signal strength. This is especially easy to do as the pre-amps, tip, lock in amp and bias ports run in multiple directions and nudging one of these cables would affect the probe. To fix this issue a 3D printed casemate was designed for the pre-amp to slide into, with a lid to prevent the pre-amp from lifting. The casemate is secured to the faraday box

using two M4 screws, which thread into the PLA casemate using two brass thread inserts.

The fragile tip harms the usability of the Kelvin probe due to the difficulty with alignment. The platinum hemisphere probe is attached to a piece of cylindrical PTFE (which interfaces with the piezo), by a small platinum wire that extends off the probe. This makes the probes position easy to later as the wire is malleable, however, this weakness caused the probe to snap off. Furthermore, the PTFE, probe interface is weak with the only connection point being a screw that friction fits into the PTFE. This also introduces a problem where the weight of the bias cable soldered onto the probe, can weigh down the PTFE sufficiently to cause it to detach from the piezo (figure 12).

To fix these issues a new probe mounting system and a redesigned probe were implemented. To begin with the new probe is made of brass with a 150nm gold coating which was evaporated onto its surface. The probes surface was increased from 2mm diameter to a 4mm diameter, increasing signal strength at the cost of spatial resolution. The new mounting system is 3D printed out of PLA, and includes a thread to interface with the piezo, and a support for the bias cable. However, if these changes fail, the probe is held up using a support that keeps the probe held in place to prevent damage to the probe. To ensure the probe holder does not come loose after prolonged usage, the PLA component is glued in place using two-part epoxy (figure 13).

The manual stage uses 3 screws to change the position of the sample plate relative to the probe. However, the stage lacks the precise stepper motors that would allow a user to maximise signal output without crashing the probe. The three screws are not easy to access

which can knock components within the faraday box, affecting probe alignment. Furthermore, the range on the Z axis of the stage is low, meaning larger samples will struggle to fit underneath. Finally, the grounding crocodile clip is difficult to attach to larger samples. To remedy this a new motorised stage is mounted within the box, on 12mm diameter steel beam. Unfortunately, X and Z motors have both become overloaded, so replacement motors should be installed, to improve operation at continuous speeds. The new stage does have a manual mode, which is easier to access and provides a smoother operating experience. Finally, the design uses a grounded metal plate, which is insulated from the rest of the stage, to ground the samples.

Appendix:

KPFM on Bruker Dimension 3100:

These instructions are mostly taken from the Nanoscope IV controller equipment manual for software version 5.12r3, chapter 8.3

- Mount the sample onto platform
- Set to tapping mode and tune the tip
- Adjust parameters to get a good tapping mode image
- In interleave controls, select (grey out):
 - Drive frequency
 - Amplitude set point
 - SPM feedback
- Set:
 - Drive amplitude to 2000mV
 - Drive phase to 0°
 - Lift start height to 0nm
 - Lift scan height to 100nm
 - Input feedback to potential
 - Interleave mode to lift
 - Scan line to interleave
 - Data type to potential
 - Scan line direction to retrace
 - AC+DC bias to tip
 - Sample bias control to ground
 - Input Igain to 0.1
 - Input Pgain to 1
 - Real time and offline plane fit to none
- Engage to tip
- Optimise Input Igain, Pgain and Lift scan height for a good image

Tables of results:

Pre-Prepared Polymer Films:

Material	Work Function (eV)	Uncertainty (eV)
PVA1 (1)	5.0	0.1
PVA1 (2)	4.7	0.1
PVA1 (3)	4.8	0.1
PS (1)	5.0	0.1
PS (2)	4.9	0.1
PS (3)	4.9	0.1
PTFE	5.3	0.1
PFA	4.8	0.1
FEP	6.1	0.1
PE	4.29	0.07
E-TFE	4.50	0.08
E-CTFE	3.72	0.08

Table 2:
Work function measurements of pre-prepared polymer films

Images:

Roughness of Spin Coated layers:

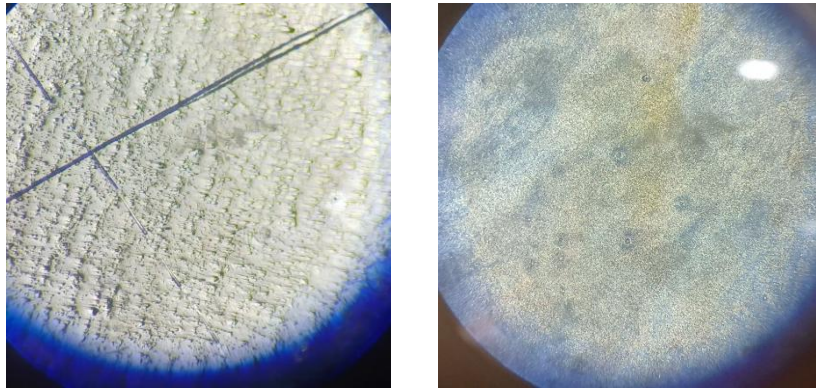


Figure 8:

Images of PVC spin coated onto smoothed steel pucks. 1 layer (left) and 5 layers (right). More layers lead to a rougher surface

Effect of cantilever on measurement of calibration grid:

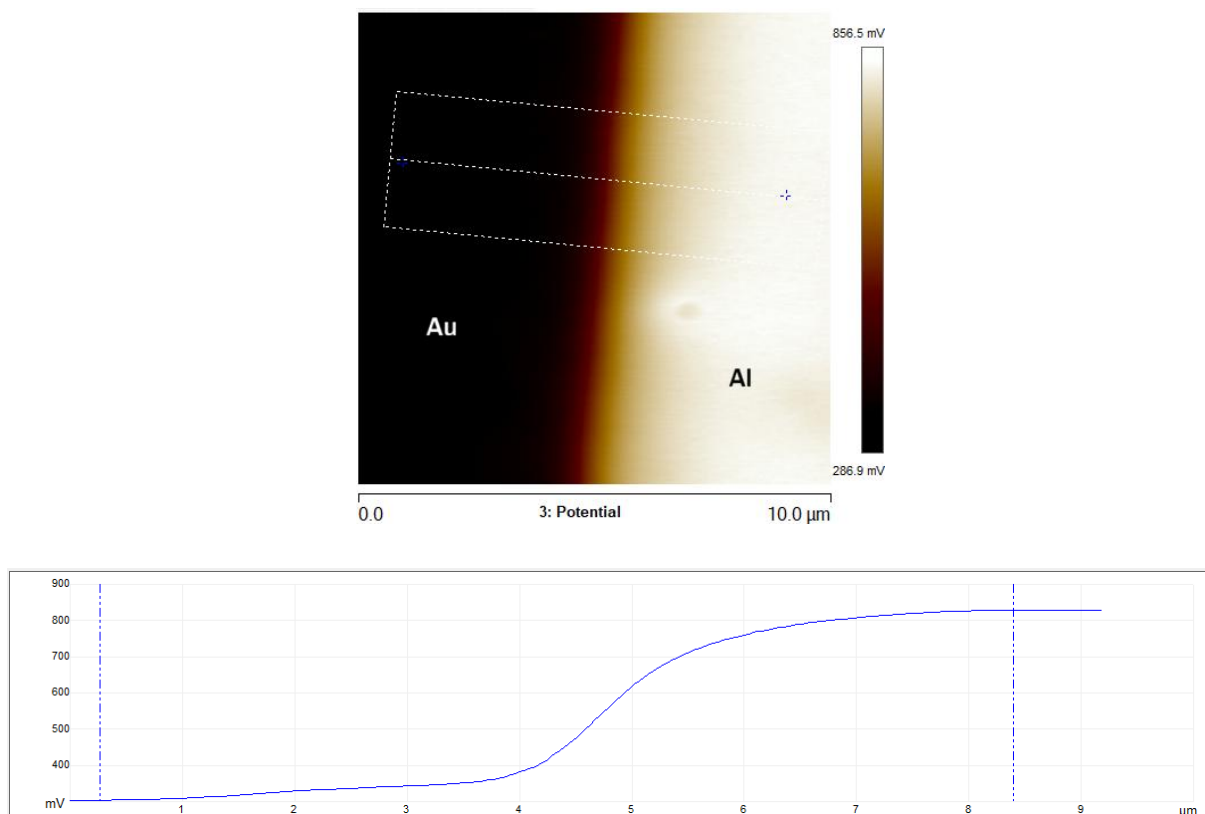


Figure 9:

CPD measurement on calibration grid. Due to the effects of the cantilever, the cantilever must be aligned parallel to the border between gold and aluminium on the calibration grid to provide accurate measurements.

FEP Charging:

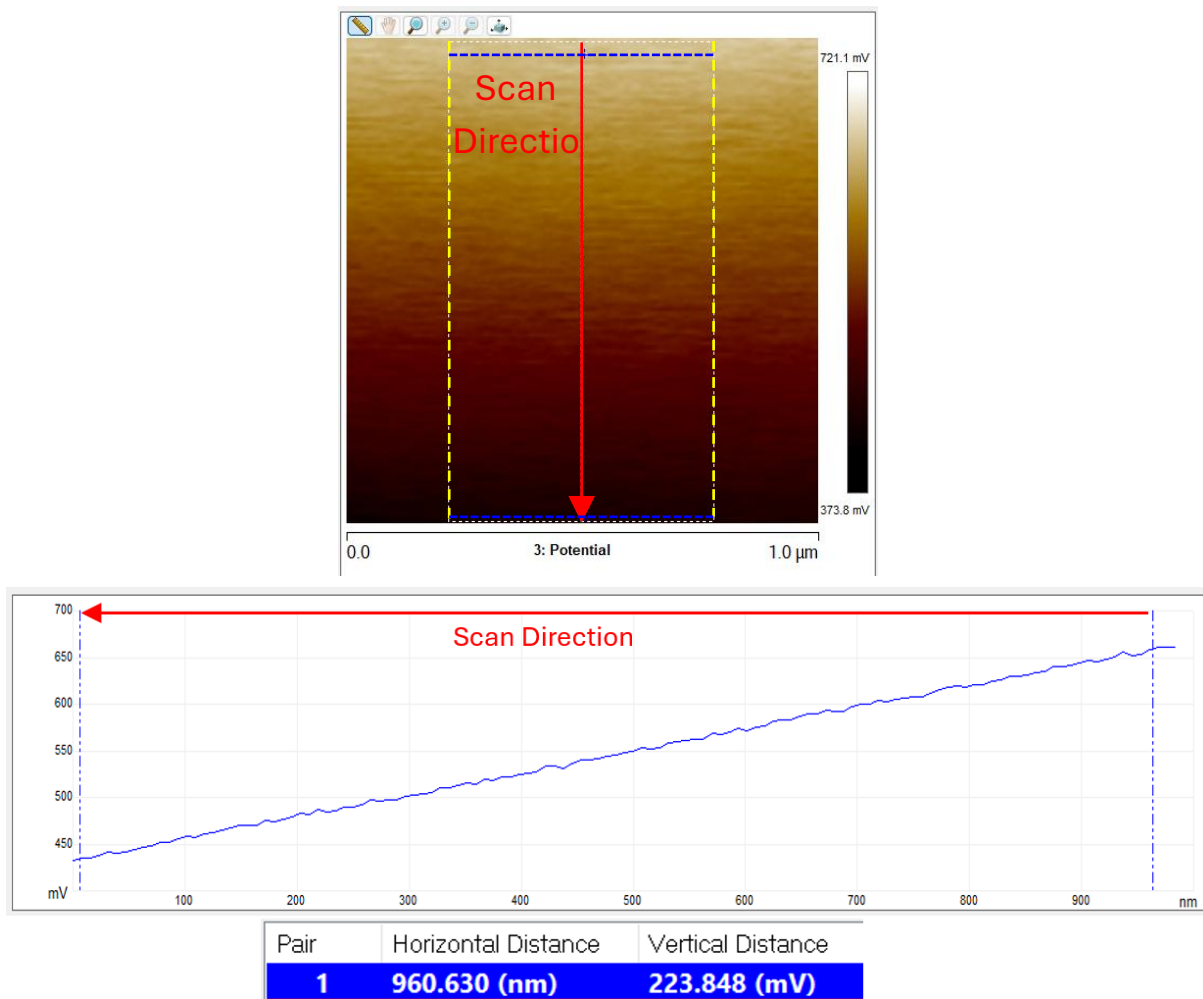


Figure 10:
Charging of FEP over time

Kelvin Probe:

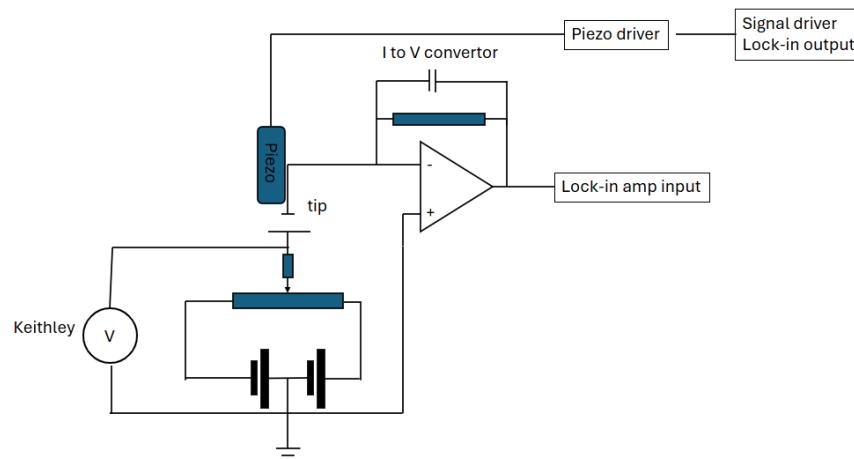


Figure 11:
Kelvin Probe circuit diagram (6)



Figure 12:
Previous piezo arm design

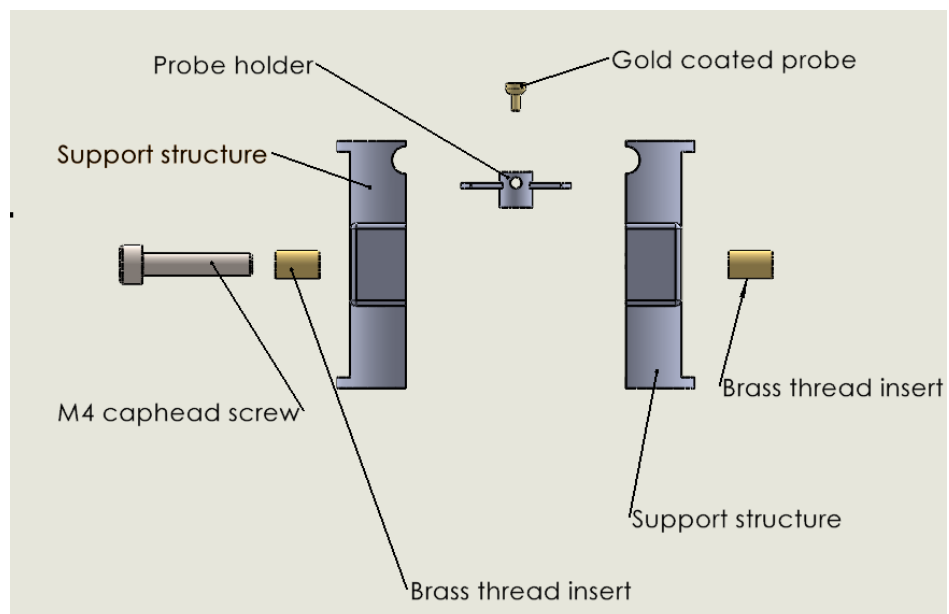


Figure 13:
New probe design

References:

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